Assignment- 1

2303A51672

Batch- 23

Privacy Risk Baseline in an AI Dataset

Scenario:

An organization wants to understand how personal or sensitive information can be exposed during AI development.

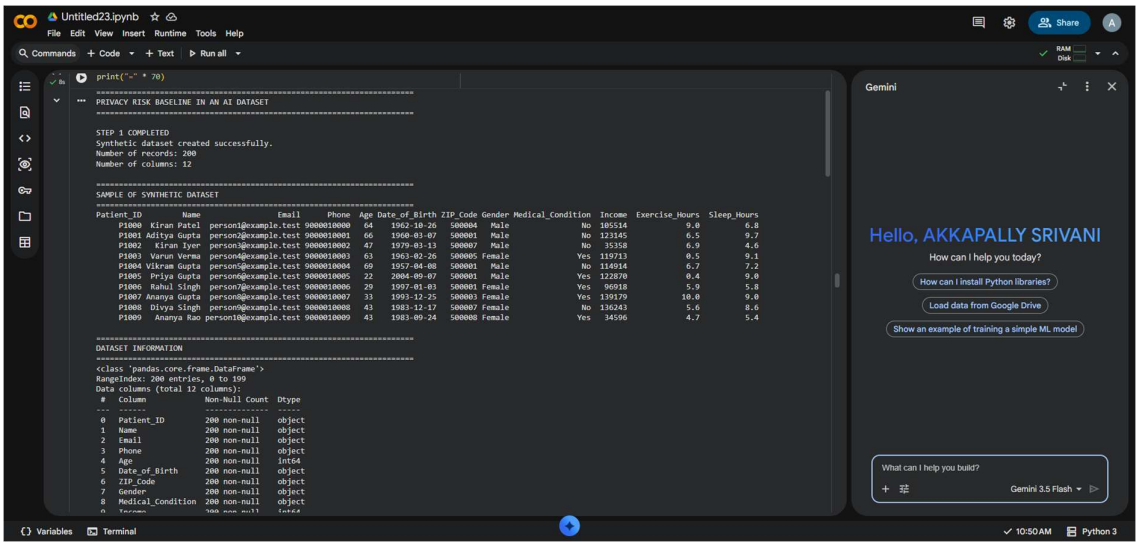
Task:

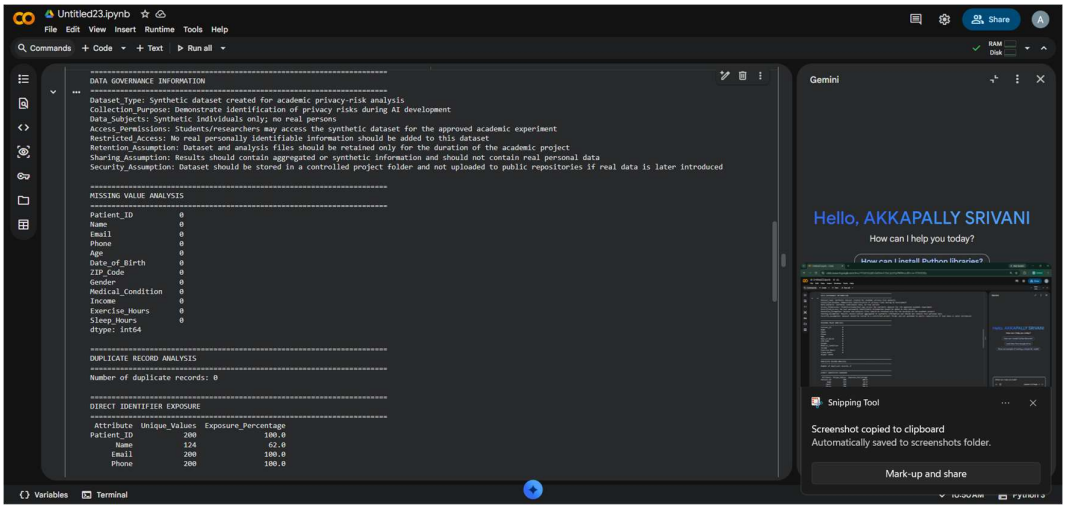
• Select an instructor-approved dataset containing identifiable or sensitive attributes, or a synthetic privacy-risk dataset.
• Identify direct identifiers, quasi-identifiers and sensitive fields.
• Document collection purpose, access permissions and retention assumptions.
• Train a simple approved model or perform a controlled data-analysis task.
• Identify possible privacy leakage paths through data, logs and outputs.
•Prepare a baseline privacy-risk table.
• Capture screenshots of the dataset and analysis.

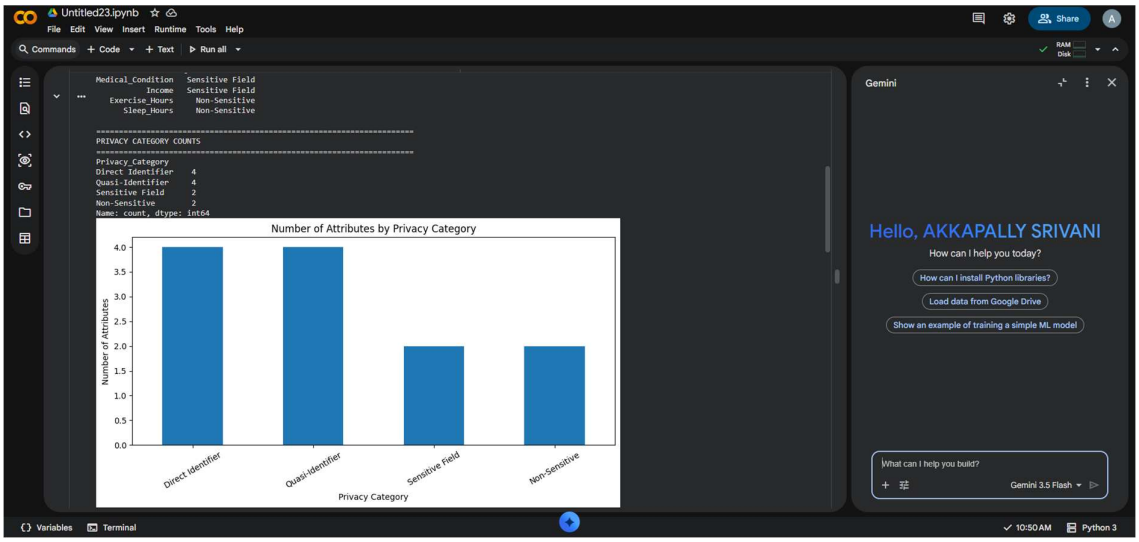
Focus:

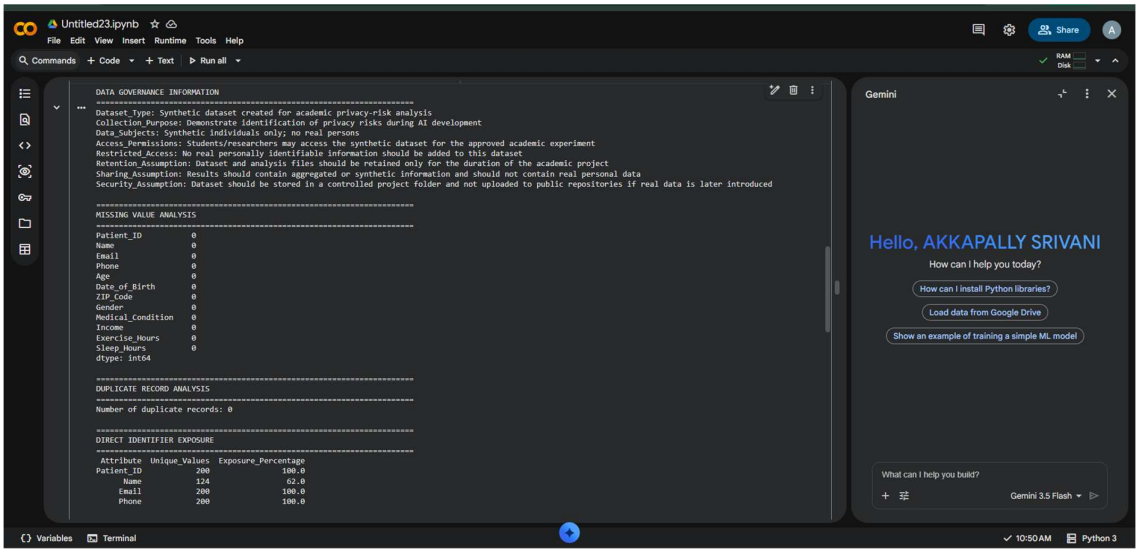
Identifying privacy risks before applying privacy-preserving techniques.

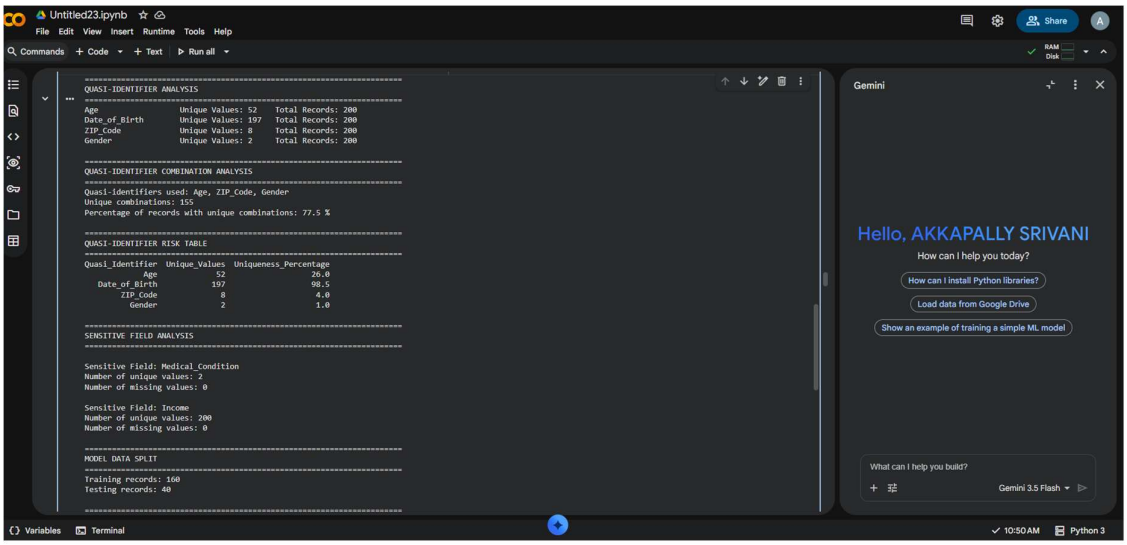
Output:

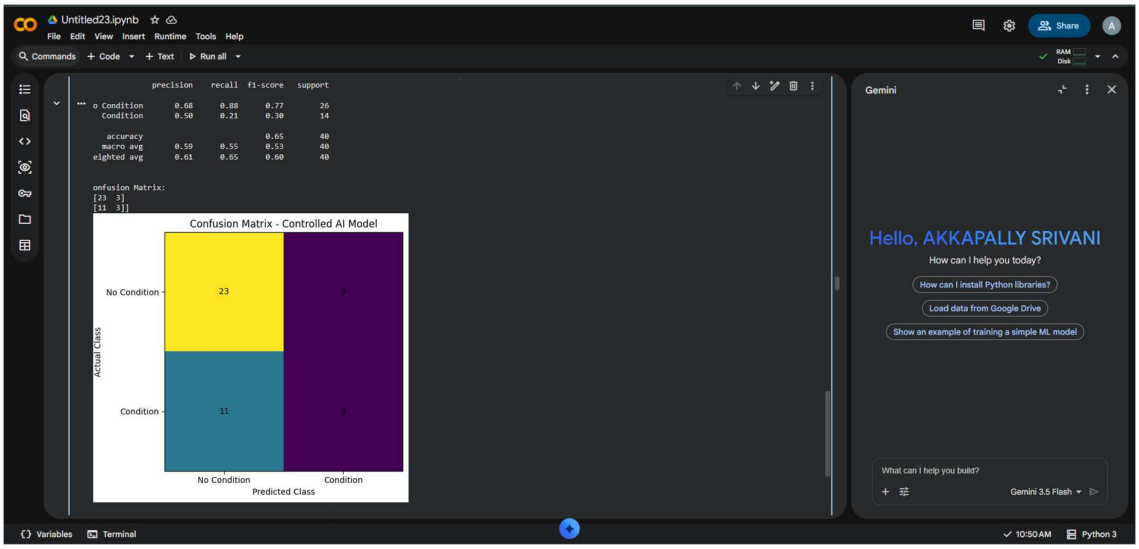


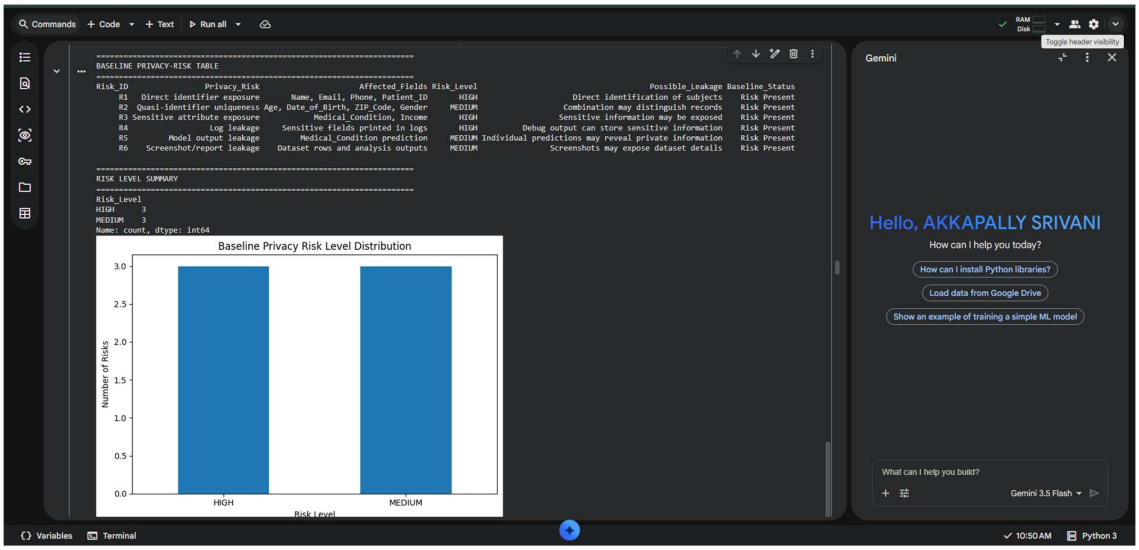

Untitled23.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Share
RAM
Disk
DATA GOVERNANCE INFORMATION
Dataset_Type: Synthetic dataset created for academic privacy-risk analysis
Collection_Purpose: Demonstrate identification of privacy risks during AI development
Data_Subjects: Synthetic individuals only; no real persons
Access_Permissions: Students/researchers may access the synthetic dataset for the approved academic experiment
Restricted_Access: No real personally identifiable information should be added to this dataset
Retention_Assumption: Dataset and analysis files should be retained only for the duration of the academic project
Sharing_Assumption: Results should contain aggregated or synthetic information and should not contain real personal data
Security_Assumption: Dataset should be stored in a controlled project folder and not uploaded to public repositories if real data is later introduced
MISSING VALUE ANALYSIS
Patient_ID 0
Name 0
Email 0
Phone 0
Age 0
Date_of_Birth 0
ZIP_Code 0
Gender 0
Medical_Condition 0
Income 0
Exercise_Hours 0
Sleep_Hours 0
dtype: int64
DUPLICATE RECORD ANALYSIS
Number of duplicate records: 0
DIRECT IDENTIFIER EXPOSURE
Attribute Unique_Values Exposure_Percentage
Patient_ID 200 100.0
Name 124 62.0
Email 200 100.0
Phone 200 100.0
Gemini
Hello, AKKAPALLY SRIVANI
How can I help you today?
Snipping Tool
Screenshot copied to clipboard
Automatically saved to screenshots folder.
Mark-up and share
Variables Terminal

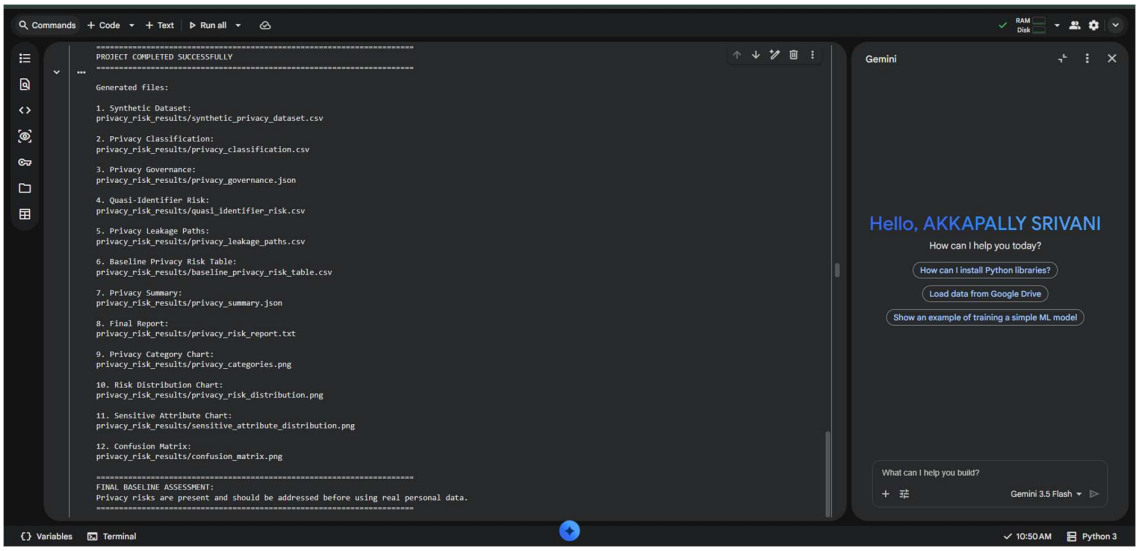

Untitled23.ipynb
File Edit View Insert Runtime Tools Help
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Share
Medical_Condition Sensitive Field
Income Sensitive Field
Exercise_Hours Non-Sensitive
Sleep_Hours Non-Sensitive
PRIVACY CATEGORY COUNTS
Privacy_Category
Direct Identifier 4
Quasi-Identifier 4
Sensitive Field 2
Non-Sensitive 2
Name: count, dtype: int64
Number of Attributes by Privacy Category
Number of Attributes
Privacy Category
Direct Identifier
Quasi-Identifier
Sensitive Field
Non-Sensitive
Gemini
Hello, AKKAPALLY SRIVANI
How can I help you today?
How can I install Python libraries?
Load data from Google Drive
Show an example of training a simple ML model
What can I help you build?
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Variables Terminal
10:50 AM
Python 3


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Name 0
Email 0
Phone 0
Age 0
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dtype: int64
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QUASI-IDENTIFIER ANALYSIS
Age Unique Values: 52 Total Records: 200
Date_of_Birth Unique Values: 197 Total Records: 200
ZIP_Code Unique Values: 8 Total Records: 200
Gender Unique Values: 2 Total Records: 200
QUASI-IDENTIFIER COMBINATION ANALYSIS
Quasi-identifiers used: Age, ZIP_Code, Gender
Unique combinations: 155
Percentage of records with unique combinations: 77.5 %
QUASI-IDENTIFIER RISK TABLE
Quasi_Identifier Unique_Values Uniqueness_Percentage
Age 52 26.0
Date_of_Birth 197 98.5
ZIP_Code 8 4.0
Gender 2 1.0
SENSITIVE FIELD ANALYSIS
Sensitive Field: Medical_Condition
Number of unique values: 2
Number of missing values: 0
Sensitive Field: Income
Number of unique values: 200
Number of missing values: 0
MODEL DATA SPLIT
Training records: 160
Testing records: 40
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File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
precision recall f1-score support
o Condition 0.68 0.88 0.77 26
Condition 0.50 0.21 0.30 14
accuracy 0.65 40
macro avg 0.59 0.55 0.53 40
eighted avg 0.61 0.65 0.60 40
onfusion Matrix:
[23 3]
[11 3]]
Confusion Matrix - Controlled AI Model
No Condition
Condition
Actual Class
23
3
11
3
No Condition
Condition
Predicted Class
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Toggle header visibility
BASELINE PRIVACY-RISK TABLE
Risk_ID Privacy_Risk Affected_Fields Risk_Level Possible_Leakage Baseline_Status
R1 Direct identifier exposure Name, Email, Phone, Patient_ID HIGH Direct identification of subjects Risk Present
R2 Quasi-identifier uniqueness Age, Date_of_Birth, ZIP_Code, Gender MEDIUM Combination may distinguish records Risk Present
R3 Sensitive attribute exposure Medical_Condition, Income HIGH Sensitive information may be exposed Risk Present
R4 Log leakage Sensitive fields printed in logs HIGH Debug output can store sensitive information Risk Present
R5 Model output leakage Medical_Condition prediction MEDIUM Individual predictions may reveal private information Risk Present
R6 Screenshot/report leakage Dataset rows and analysis outputs MEDIUM Screenshots may expose dataset details Risk Present
RISK LEVEL SUMMARY
Risk_Level
HIGH 3
MEDIUM 3
Name: count, dtype: int64
Baseline Privacy Risk Level Distribution
Number of Risks
3.0
2.5
2.0
1.5
1.0
0.5
0.0
HIGH
MEDIUM
Risk Level
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Commands
Code
Text
Run all
RAM
Disk
PROJECT COMPLETED SUCCESSFULLY
Generated files:
1. Synthetic Dataset:
privacy_risk_results/synthetic_privacy_dataset.csv
2. Privacy Classification:
privacy_risk_results/privacy_classification.csv
3. Privacy Governance:
privacy_risk_results/privacy_governance.json
4. Quasi-Identifier Risk:
privacy_risk_results/quasi_identifier_risk.csv
5. Privacy Leakage Paths:
privacy_risk_results/privacy_leakage_paths.csv
6. Baseline Privacy Risk Table:
privacy_risk_results/baseline_privacy_risk_table.csv
7. Privacy Summary:
privacy_risk_results/privacy_summary.json
8. Final Report:
privacy_risk_results/privacy_risk_report.txt
9. Privacy Category Chart:
privacy_risk_results/privacy_categories.png
10. Risk Distribution Chart:
privacy_risk_results/privacy_risk_distribution.png
11. Sensitive Attribute Chart:
privacy_risk_results/sensitive_attribute_distribution.png
12. Confusion Matrix:
privacy_risk_results/confusion_matrix.png
FINAL BASELINE ASSESSMENT:
Privacy risks are present and should be addressed before using real personal data.
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